MILITARY SPECIFICATION

FUEL OIL, DIESEL, MARINE

This amendment forms a part of Military Specification MIL-F-16884G, dated 7 March 1973, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.2: Under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)", add:

"D 1401 - Emulsion Characteristics of Petroleum Oils and Synthetic Fluids, Test For."

PAGE 2

TABLE I: Delete and substitute:

TABLE I. Chemical and physical requirements.

Characteristics	Requirements	FED-STD-791 test method	ASTM test method
Ignition quality, cetane number (min) (see 4.6.1)	45		D 613
Appearance[1/]	Clear, bright, and free from visible particulate matter.		
Distillation:			
90 percent point, °C (°F) (max)	357°C (675°F)		D 86
End point, °C (°F) (max)[2/]	385°C (725°F)		
Residue plus loss, percent (max)	3.0		
Flash point, °C (°F) (min)	60°C (140°F)		D 93
Pour point, °C (°F) (max)	-6.7°C (20°F)		D 97
Cloud point, °C (°F) (max)	-1.1°C (30°F)		D 2500
Viscosity at 37.8°C (100°F) Kinematic, centistokes	1.8 to 4.5		D 445
Carbon residue, on 10 percent bottoms, percent (max) (see 4.6.2)	0.20		D 524
Sulfur, percent (max)	1.00		D 129[3/]
Corrosion (max) at 100°C (212°F)	No. 1 ASTM		D 130
Color (max)	3		D 1500
Ash, percent (max)	0.005		D 482
Gravity (hydrometer)	Record		D 1298[4/]
Demulsification, minutes (max) (see 4.6.3)	10		D 1401
Acid number (max)	0.30		D 974
Neutrality	Neutral	5101	------→
Aniline point, °C (°F)	Record		D 611
Accelerated stability, total insolubles mg/100 mL (max)	2.5[5/]		D 2274

1/ A slight haze is acceptable providing a maximum water and sediment of 0.01 percent is obtained using procedure ASTM D 2709.

2/ As the end point of the distillation is approached, if either a thermometer reading 725°F (385°C) or a decomposition point is observed, discontinue the heating and resume the procedure as directed in ASTM D 86.

3/ ASTM D 1552 and ASTM D 2622 may be used as alternative methods.

4/ ASTM D 287 may be used as an alternative method.

5/ Average of [illegible] determinations accepted.

FSC 9140

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PAGE 3

3.4.1(f): Delete "25 percent min." and substitute "25 percent max."

PAGE 4

4.6.3, second line: Delete "method 3201 of FED-STD-791", and substitute "ASTM D 1401".

4.6.3(a): Add to end of sentence:

"(see note 3, paragraph 6, ASTM D 1401 procedure)."

Custodians:
Army - MR
Navy - SH
Air Force - 68
Review activities:
Navy - YD
User activities:
Navy - CG, MC

Preparing activity:
Navy - SH
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